

Modulation of Phonetic Realizations in Cantonese Dialogue with Human and AI Interlocutors

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Abstract

Speech patterns in human-AI interaction have received much attention recently. Prior research has focused on emotionally expressive systems in intonation languages. Less is known about how interlocutor identity (human vs. AI) shapes phonetic realizations in dialogue in tone languages. This study examined whether interacting with a human versus an AI interlocutor modulates phonetic properties in Cantonese. Scripted interactive dialogues in neutral and negative contexts with human and AI interlocutors in a Wizard-of-Oz setup were collected from 17 native Cantonese speakers. Word duration, F0 of the six tones and F1 and F2 of three corner vowels were measured. Results show evidence of phonetic modulation in AI interaction with shorter target word duration and specific tonal difference compared to human interaction. Emotional effect was marginal with a larger vowel space area in the negative context. The findings indicate more fine-grained interlocutor effects than previously reported.

Index Terms: phonetic realization, human-AI interaction, emotion, Cantonese

1. Introduction

Speech modulation refers to the tendency for interlocutors to adjust their speech based on the characteristics of other interlocutors. This adaptive behaviour has been documented at multiple linguistic levels, such as phonetic and syntactic realizations, and has been widely observed in human-human interaction across different types of interaction such as child-directed (e.g., [1]), adult-directed (e.g., [2]), foreigner-directed (e.g., [3]) speech. Beyond these types of interactions, recently, with the rapid developments in the technology of Artificial Intelligence (AI), human began communicating with AI-generated speech products (e.g., Alexa). A new type of interaction has emerged, namely human-AI interaction.

Prior work on phonetic modulation in human-AI interaction has largely focused on prosocial or positively expressive systems ([4], [5]). These studies demonstrated that humans modulated various aspects of their speech, such as duration and speech rate when interacting with AI interlocutors. However, recent proposals concerning ‘antagonistic AI’ [6] have drawn attention to synthetic agents that adopt less cooperative or even confrontational communicative styles. While such designs can make the agents or AI voice more diversified and more human-like, little is known about how users react to these negative AI expressions. Although [4] and [5] included a small number of negatively valenced single-word interjections (e.g., darn), the effect of negative emotion was not analyzed in detail. As a result, it remains unclear how humans modulate their speech to negatively expressive or ‘antagonistic’ AI speech, highlighting a significant research gap. In view of this, this study examines the effect of emotion on human-AI interaction with a specific

focus on negative emotion.

At the same time, research on phonetic modulation has shown that task type matters (e.g., [7]). While the shadowing paradigm reveals rapid imitation effects, interactive dialogue tasks may elicit more subtle and temporally localized adjustments. As relatively less attention has been paid to how the identity of interlocutors (human vs. AI) shaped fine-grained phonetic realization in interactive settings, it is unclear whether interacting with a synthetic interlocutor would systematically influence phonetic properties of speech beyond global adjustments. Scripted dialogue paradigm provides an effective balance between ecological validity and experimental control, allowing investigation of conversational modulation under tightly controlled lexical content. Previous research using scripted dialogue tasks has demonstrated differences in the modulation of speech features between tasks, such as F0, intensity, and vowel formants (e.g., [8]), supporting its feasibility for investigating phonetic modulation under controlled yet interactive condition.

The present study adopted the scripted dialogue paradigm to examine whether interlocutor identity (Human vs. AI) and emotional stance (Neutral vs. Negative) modulate phonetic realizations in Hong Kong Cantonese. Specifically, we investigated acoustic measures including utterance speech rate, duration of the target words, time-normalized F0 trajectories of the lexical tones and the first two formants (F1 and F2) of the target vowels. Besides, emotional stance was manipulated within dialogue to explore how negative emotion modulates acoustic realization in interaction.

Very few previous studies on human-AI interaction have investigated tone languages, as they tended to focus on intonation languages (e.g., English in [8], [9], [10], and German in [7], [11], [12]). Cantonese, a tone language with six contrastive lexical tones, provides a particularly suitable testing ground due to its rich tonal system, which includes three level (high-level, 55; mid-level, 33; low-level, 22), two rising (high-rising, 25; low-rising, 23), and one falling (low-falling, 21) tones. As part of a larger study on phonetic accommodation of Cantonese sound change features, some target words of this project in this study consist of the segmental sound change features. Nevertheless, as this study investigated the acoustic aspects of speech modulation, sound change is not the focus.

This exploratory study asked two main research questions: (1) Does interacting with an AI versus a human interlocutor influence fine-grained acoustic realizations in Cantonese? (2) Does emotional stance further modulate these interlocutor-related effects? By examining acoustic measures including duration, fundamental frequency (F0) and vowel formants (F1 and F2) in interactive speech, this study aimed to expand our understanding of how interlocutor identity shapes phonetic realizations in human-AI communication. Despite being exploratory, based on insights from previous work, we hypothesized that interlocutor identity would exhibit an effect

on the phonetic realizations of Cantonese globally, with additional fine-grained, localized impacts depending on specific lexical tones, which is further modulated by the emotional stance.

2. Method

The study employed a within-subject design manipulating Interlocutor Identity (Human vs. AI) and Emotional Stance (Neutral vs. Negative) to examine whether participants responded to emotionally expressive speech differently and exhibited different acoustic patterns between interlocutor identities.

2.1. Participants

Seventeen native Hong Kong Cantonese speakers participated in this experiment (9M, 8F; aged 18-27, $M = 21.17$). They reported having normal hearing and no history of speech disorders or cognitive disabilities. Upon arrival, participants provided informed consent and completed a language-background questionnaire. The experiment was conducted in a sound-proof booth. The participants were compensated for their participation in the experiment.

2.2. Materials and stimuli

2.2.1. Dialogue construction

The experiment used a scripted dialogue task based on the Wizard-of-Oz paradigm [13] (see more in Section 2.3). Similar to the alternative reading task in [14], [15], participants first read their scripted dialogue turns aloud to familiarize themselves with the content. They then engaged in an interactive phase with pre-recorded audio-video stimuli of either a human or an AI model talker.

Each dialogue consisted of six alternating turns (three by the model talker and three by the participant). Across the experiment, there are 20 dialogues. The target word occurred six times in each dialogue: twice in the model talker (1st and 5th turn); four times in the participants (2nd and 6th turn). The target words are 中午 (/tsuŋ⁵⁵ ŋ²³/, ‘noon’), 五個 (/ŋ²³ kɔː³³/, ‘five items’), 誤會 (/ŋ²³ wuːi²²/, ‘misunderstanding’), 領悟 (/lŋ²³ ŋ²²/, ‘understand’), 理想 (/lei²³ sɔːŋ²⁵/, ‘ideal’), 上車 (/sɔːŋ²³ tsheː⁵⁵/, ‘buy a property’), 影相 (/jŋ²⁵ sɔːŋ²⁵/, ‘take a photo’), 商務 (/sɔːŋ⁵⁵ mou²²/, ‘business’), 尼龍 (/nei²¹ lŋ²¹/, ‘nylon’), 普洱 (/p^hou²⁵ nei²⁵/, ‘Pu’er tea’), 意外 (/jiː³³ ŋɔːi²²/, ‘accident’), 愛心 (/ɔːi³³ sɛm⁵⁵/, ‘care’), 曖昧 (/ɔːi²⁵ muːi²²/, ‘ambiguous’), 睇書 (/t^hei²⁵ syː⁵⁵/, ‘read a book’), 暑假 (/syː²⁵ kaː³³/, ‘summer holiday’), 薯條 (/syː²¹ t^hiːu²¹/, ‘French fries’), 種樹 (/tsuŋ³³ syː²²/, ‘plant a tree’), 肥膩 (/fei²¹ nei²²/, ‘greasy’), 癡呆 /ts^hiː⁵⁵ ŋɔːi²¹/, ‘Alzheimer’s’), and 餌塊 (/nei²² faːi³³/, ‘rice cake’).

As the example shown in Table 1, in participant(B)’s utterances, the target disyllabic word 中午 (/tsuŋ⁵⁵ ŋ²³/, ‘noon’), highlighted in bold, was always positioned medially to minimize potential prosodic boundary effects. In total, each participant was expected to produce 80 utterances (4 turns x 20 dialogues).

2.2.2. Emotion manipulation

Two emotional conditions were implemented: neutral and negative. Each dialogue began with two neutral turns introducing the topic. Starting from the 3rd turn, the model talker shifted to a negative stance (e.g., blame, complaint, or criticism), and the participant responded accordingly until the end of that dialogue. Emotion was conveyed through both dialogue context, facial expression and natural prosody expressing criticism or dissatisfaction. Target words appeared in both neutral and negative turns of the dialogue equally, allowing comparison across emotional contexts. As illustrated in Table 1, the target word 中午 (/tsuŋ⁵⁵ ŋ²³/, ‘noon’) occurred

in both stances twice. Also, each turn only consisted of 2 to 3 short utterances with fewer than 20 Chinese characters. This was specifically designed to balance production ease, natural conversational flow, and token comparability.

2.2.3. Human and AI model talkers

In the human condition, a female native Cantonese speaker recorded both audio and video stimuli using a SONY video recorder. The talker faced the camera in the front and maintained eye contact to simulate a real-time video call. In the AI condition, the visual stimuli of an avatar were generated using iPhone Memoji [16] to convey facial gestures associated with negative emotion. Recordings in both human and AI conditions were done simultaneously. To avoid the pitfalls of previous studies on using existing AI assistants which voices are different from the human interlocutors in the experiments, the present study developed a customized TTS voice for the AI model talker trained on approximately 3000 Cantonese sentences recorded by the same female speaker used in the human condition, using a modified version of the duration-informed DurlAN [17] model, which takes phoneme duration, pitch, and energy as input parameters. These features were extracted from the original human recordings and used as input to control the synthesis process. This allowed the synthesized speech to closely replicate the prosody, and rhythm of the original recordings while preserving the unique pronunciation style and voice quality of the model interlocutors. The generated AI voice preserved key speaking characteristics of the original speaker while exhibiting a slightly mechanical quality due to reduced resolution and timbral differences. The synthesized audio replaced the original audio track in the AI videos. This procedure ensured comparable speech content, posture, and facial orientation across human and AI conditions.

Table 1: Structure of a scripted dialogue.

A	中午喇，準備去食飯未呀？[NEU] (It’s noon now, are you ready to go eat yet?)
B	差唔多喇，我諗 中午 度度都會好多人，咁你 中午 諗住食咩呀？(Almost, I think everywhere will be crowded at noon , so what are you thinking of eating at noon ?)
A	你問嚟做咩啫？次次都做跟尾狗，煩唔煩㗎嘅！[NEG] (Why are you asking? You always follow along like a tailwagging dog. Don’t you find that annoying?)
B	使唔使噉講嘢啊，我只係想參考下你嘅意見啫。(Do you need to talk like that? I just wanted to get your opinion.)
A	日日都問，係咪以後 中午 食咩都要同你匯報啊？[NEG] (You ask that every day. Do I need to report to you what I eat at noon from now on?)
B	乜你 中午 食咩都咁神秘，以後 中午 都唔問你囉！(Why is what you eat at noon so secretive? I won’t ask what you’re having at noon anymore then!)
Note: A = Model talker, B = Participant; NEU = Neutral emotion, NEG = Negative emotion	

2.3. Procedure

The experiment was implemented in E-Prime 3.0 software (Psychology Software Tools, Pittsburgh, PA) [18]. Participants completed two practice dialogues with neutral content. Then they completed 20 experimental dialogues: 10 with the human model talker and 10 with the AI model talker. The interlocutor order was counterbalanced across participants. An on-screen message indicated when the dialogue interlocutor switched.

The experiment followed a Wizard-of-Oz setup [13]. The experimenter, located in a neighboring room, controlled the timing of the dialogue turns while monitoring participants' speech in real time via Zoom audio [19]. They were informed that the audio would be used solely for communication purposes, allowing them to contact the experimenter in case of unexpected issues (e.g., technical errors). Yet, they were not informed about the Wizard-of-Oz setup.

2.4. Acoustic and statistical analysis

Acoustic analyses were conducted in Praat [20]. Duration of the target words and character and utterance-level speech rate were analyzed using linear mixed-effects models implemented in lme4 [21] in R [22]. Fixed effects included Interlocutor Identity (Human vs. AI) and Emotional stance (Neutral vs. Negative). Random intercepts for Participant and Dialogue were included to account for repeated measures. Model parameters were estimated using restricted maximum likelihood (REML). F0 trajectories of the target words were analyzed using generalized additive mixed models (GAMMs) implemented in mgcv, using the bam() function [23]. The dependent variable was speaker-normalized zF0 sampled at ten normalized time points per syllable. Non-linear contour shapes were modeled using smooth terms over normalized time, with separate smooths for each Interlocutor Identity $\times$ Emotion $\times$ Tone Category. This study included all six tones in Cantonese. Random intercepts for Participant and Token were included.

3. Data processing and analysis

Audio recordings were first force-aligned using Montreal Forced Aligner (MFA) [24] and subsequently manually corrected by phonetically-trained native Cantonese speakers. The corrected data were segmented at multiple hierarchical levels: (1) utterance level (for each utterance containing a noticeable pause), (2) syllable level for each Chinese character, (3) target-word level (disyllabic target words, with separate intervals for each syllable), and (4) sonorant level within each syllable and (5) vowel level. Segmented data were processed using ProsodyPro [25] for data extraction.

3.1. Durational analysis

Acoustic measures included utterance-level speech rate (syllables per second), target word duration (ms, e.g., 中 午 /tsɒŋ⁵⁵ ɳ²³/, 'noon'), and target character duration (ms, e.g., 中 /tsɒŋ⁵⁵/, 'middle'). The target disyllabic word duration included short pause between two target characters rather than a simple combination of the two target monosyllables. Linear mixed-effects models were fitted with Interlocutor Identity (Human vs. AI) and Emotional stance (Neutral vs. Negative) as fixed effects. Random intercepts were included for Participant and Dialogue to account for repeated measures.

Table 2: Duration of target words.

Target type	Interlocutor	Emotion	Mean duration (ms)
Disyllabic Word	Human	Neutral	381±97.6
		Negative	380±91.0
	AI	Neutral	370±84.8
		Negative	375±86.8
Syllable	Human	Neutral	194±58.0
		Negative	194±62.4
	AI	Neutral	187±51.7
		Negative	191±56.8

The linear mixed effect model revealed a significant main effect of Interlocutor Identity ($\beta = -10.891$, $p = .04$) for the

duration of the target disyllabic word, indicating shorter durations in the AI condition relative to the Human condition. No significant main effect of Emotion or interaction between Interlocutor Identity and Emotion was found. Table 2 presents the means of the duration. For target syllable duration, the mixed-effects model revealed a marginal main effect of Interlocutor Identity ($\beta = -5.986$, $p = .06$), again with a trend of shorter durations in the AI condition. For utterance-level speech rate, no significant effects of Interlocutor Identity or Emotion were found. This indicates that the Interlocutor effect on duration was mainly located in the target words rather than as a global speech rate phenomenon.

3.2. F0 trajectory analysis

F0 contours were analyzed at the level of tone-bearing syllables (target syllables). For each syllable, F0 was sampled at ten equally spaced normalized time points (0–100%) across the sonorant interval. To reduce inter-speaker differences in pitch range, F0 values were z-normalized within speaker to enable cross-speaker comparison. Time-varying F0 trajectories were modeled using generalized additive mixed models (GAMMs) ([26], [27]). The dependent variable was normalized F0, while the Tone Category (6 Tones) $\times$ Interlocutor Identity (Human vs. AI) $\times$ Emotion (Negative vs. Neutral) was fitted as the predictors. Smooth terms modeled the F0 trajectory over time, including the above-mentioned predictor-specific smooths and participant-specific random smooths. An AR(1) autocorrelation structure was included.

Time-localized differences between Interlocutor Identity and Emotion across six tones were assessed using the plot_diff function in the itsadug package [28]. Time intervals in which two curves were significantly different were identified when the 95% confidence interval of the difference did not include zero.

For the current set of participants ($n = 17$), the Interlocutor Identity and emotion effects were tone-specific and temporally localized. As shown in Figure 1, for Tone 1 (significant difference interval: 2.55 ~ 7.73), Tone 4 (significant difference interval: 1 ~ 5) under neutral emotion and Tone 4 under negative emotion (significant difference interval: 1.18 ~ 8.36), participants' productions toward an AI interlocutor exhibited a significantly raised F0 contour compared to a Human interlocutor. Conversely, for Tone 5 (significant difference interval: 4 ~ 8.45) under neutral emotion, participants' production toward an AI interlocutor exhibited a significantly lowered F0 contour compared to a Human interlocutor. Emotion effect was observed in Tone 2 only. Participants' production under negative emotion exhibited a significantly raised F0 contour compared to neutral emotion for both interlocutor identities (significant difference interval: AI: 1.36 ~ 10; Human: 1.45~10).

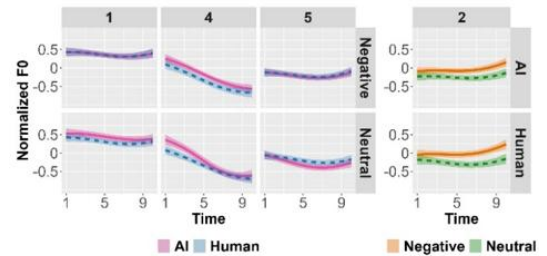


Figure 1: Smoothed curves of the tone contours produced contrasted by different interlocutors (right panels) and by different emotions (left panels).

3.3. Vowel space analysis

Six target words containing the corner vowels /i/ /a:/ /o/ were analyzed for the vowel space analysis. These vowels were from the target words so they appeared in both emotions equally but

not Interlocutor Identity. Therefore, we restricted our analysis of the effect of emotion only to avoid any potential overgeneralization caused by imbalanced data. Due to the lack of /a:/ vowel, one token of /a:/ was extracted from the first half of the diphthong /a:i/. The F1 and F2 values of all the other vowels were measured from the 10% to 100% portion of the vowel (excluding the initial 10% to avoid consonantal transition effects). The diphthong /a:i/ was measured using the 10% to 40% portion of the vowel.

Midpoint F1 and F2 values were averaged across tokens for each vowel category (/i:/, /a:/, /u:/) within each participant, Interlocutor Identity, and Emotion. The resulting mean formant values were converted from Hz to the ERB scale using the `normErb()` function in the `phonR` package [29] to obtain perceptually scaled vowel space representations. The three corner vowels were then used to compute the vowel space area (VSA) as the area of the triangle defined by their ERB-transformed F1–F2 coordinates. A linear mixed-effects model revealed a marginal effect of emotion on vowel space area ($\beta = 1.07$, $p = .057$), with larger VSA in negative relative to neutral emotion, as shown in Figure 2. Random intercept for participant was included to account for between-speaker variability.

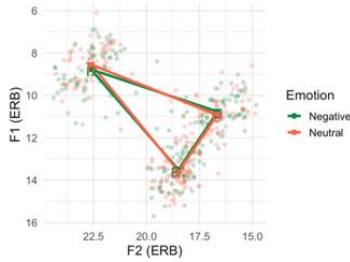


Figure 2: Vowel space by emotion. Points represent individual vowel tokens plotted in ERB-transformed F1–F2 space. Triangles connect the mean values of /i:/, /a:/, and /u:/.

4. Discussion

This study investigated whether phonetic realizations differ between human-human and human-AI scripted dialogue in the tone language Cantonese, and whether emotional stance further modulates such differences. With respect to temporal measures, overall utterance speech rate did not differ significantly between interactions with AI and human interlocutors. However, target word duration was significantly shorter in the AI condition, for both the target disyllabic words and target syllables, indicating a more selective effect on the target items than a general global change in speech rate. Given that the target words recurred multiple times throughout the dialogue, they represent a stable and reliable locus for examining the Interlocutor effect. The shorter durations observed in the AI condition may reflect more succinct speech directed toward an AI interlocutor, potentially indicating reduced elaboration or increased communicative efficiency. This pattern contrasts with previous findings reporting longer durations [5] and slower speech rates [8] in speech directed toward AI relative to human interlocutors. Further work is needed to investigate the contrary findings. One possible reason is that the voice and speech rate of the AI interlocutors in previous studies were provided by existing AI assistants (e.g. Alexa) which differed from those of the human interlocutors. The experimenters had no control over the AI voices. It is possible that speech rates might be different in the AI and Human conditions in previous studies, which could confound the findings. Nevertheless, in our study, both the voice and the speech rate were comparable in the AI and Human conditions as the AI voice was customized based on the human voice. Thus, the consistent finding of shorter target word duration in our study is robust.

In terms of F0, Tone 1 and Tone 4 were raised while Tone 5 was lowered when the participants were conversing with an AI interlocutor in a neutral emotion. Additionally, Tone 4 was also raised when conversing with an AI interlocutor in negative emotion. There was no significant F0 difference in the other tones. The results indicate that the Interlocutor effect was tone-specific rather than a general shift of F0 height. Previous studies showing an overall shift in mean F0 (e.g., a higher mean F0 in English [8] and German [11] or a lower mean F0 in English [10]) when participants interacted with AI assistants were all based on intonation languages. They did not investigate F0 patterns at the word level. The data from this study on Cantonese demonstrate finer F0 manipulation of the participants towards different interlocutors. While it needs further investigation to decipher the opposite raising and lowering effects of F0 on different tones, previous studies also showed opposite F0 changes English, as mentioned above. It is possible that the participants were enlarging the pitch range or enhancing specific tonal features. What is clear is that the participants were sensitive to the difference in interaction between human and AI interlocutors, and they could adjust their specific tonal realizations accordingly.

The results of VSA suggest that the vowel space tended to expand under a negative emotion relative to a neutral emotion. Although the effect of emotion on vowel space was marginal, probably due to the small sample size of vowel tokens, the direction and magnitude of the effect are consistent with accounts linking emotional intensity to increased articulatory dispersion. This is corroborated by the significant finding of a raised F0 contour of Tone 2 under negative emotion for both human and AI interlocutors. Our study contrasting neutral and negative emotional contexts extends the findings of [5] who examined neutral and positive emotions in human-human and human-AI interaction. They found similar responses to emotion in both human and AI interlocutors. The more general overall emotional effects in [5] compared to the isolated effects in the current study may be due to methodological differences. [5] used a shadowing task with single-word interjections to manipulate emotions, while our scripted dialogues embedded semantically neutral targets within interactional sequences involving blame, criticism, or disagreement. In our materials, emotional stance was conveyed through a more natural interpersonal context rather than lexical content. Participants may have different responses towards the negative expressions, averaging out the subtle emotional effects on phonetic realizations as a result. Further investigation of individual variation is thus warranted.

A further consideration concerns the structure of the emotional manipulation itself. Because neutral and negative utterances appeared within the same brief dialogue with only 6 turns, the rapid alternation of affect may have limited the time available for phonetic adjustments, thus minimizing the emotional effects. Modulation in speech patterns can be affected by the interlocutor’s immediately preceding turn [9], and abrupt emotional stance shifts may interrupt the incremental processes through which change accumulates. Our ongoing work includes better-designed dialogues that sustain a consistent emotional tone across turns, allowing clearer observation of emotion-driven phonetic adaptation.

In conclusion, the present study gives positive answers to the two main research questions that interlocutor identity does affect phonetic realizations in terms of duration, F0 and vowel space in Cantonese dialogue, and emotional stance further modulates these interlocutor-related effects, albeit with modest effects. Moreover, our findings on target words and specific tones demonstrate that the effects are fine-grained. Participants could control the different phonetic realizations in a word-specific manner, in addition to the overall gross patterns reported by previous studies. Nevertheless, given the relatively small effects in the present study, further investigation is needed to substantiate our findings on human-human and human-AI interactions in a tone language.

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6. Generative AI Disclosure

During the preparation of this manuscript, the authors used ChatGPT (www.chat.openai.com) for editing the text for English usage. No generative AI was used for data collection or image generation.

7. References

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